

JKBOSE

2026 CLASS 11TH

10

MOTION IN STRAIGHT LINE

LECTURE-5

PHYSICS

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Sir g



Topics To Be Covered

- 1** revision ✓
- 2** GABBAR NUMERICAL SERIES ✓
- 3**
- 4** ..

“PERFECTION IS NOT
ATTAIABLE,
BUT IF WE CHASE
PERFECTION WE CAN
CATCH EXCELLENCE

Recap From 1st Lec.

Motion

- move one place to another place with respect to

Rest

Donot move from one place to another place with respect to

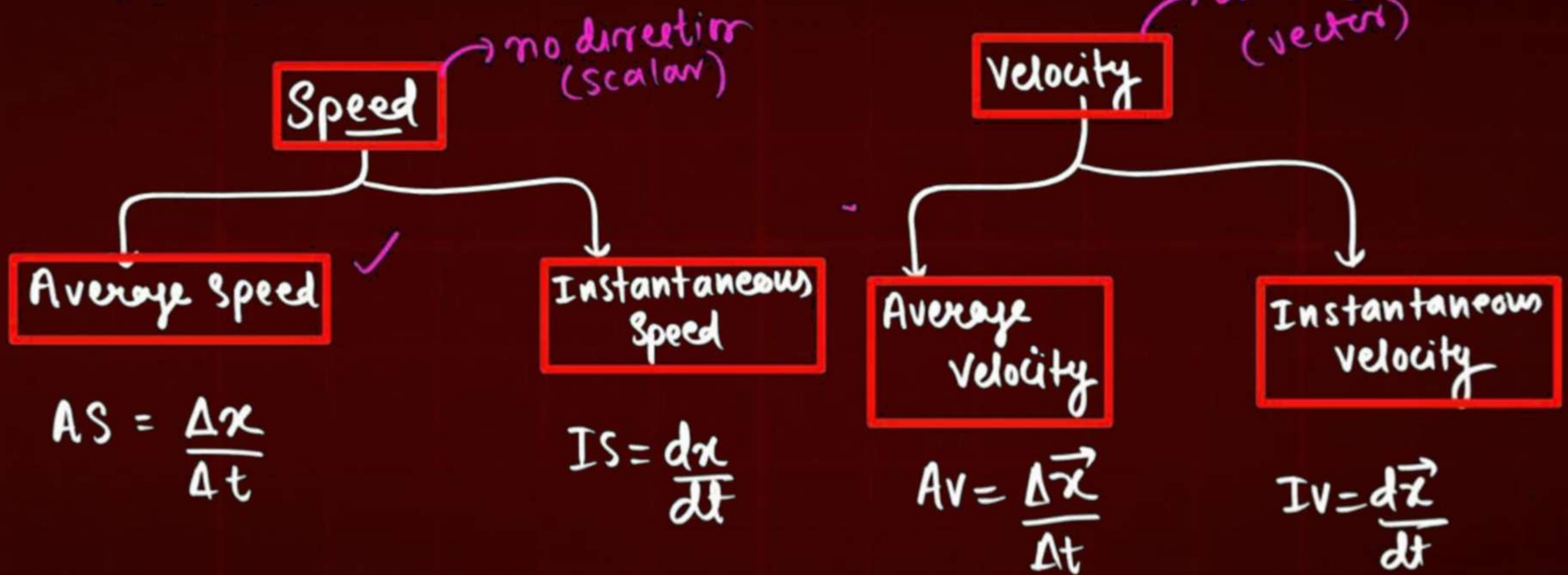
Reference point
(Motion is Relative)

$$1 \frac{\text{km}}{\text{hr}} = \frac{5}{18} \text{ m/s}$$

Numericals

distance	displacement
• Total path • metre • Scalar • Positive +	• Shortest path • metre • vector • + - zero
Speed	Velocity
• $\text{Speed} = \frac{\text{distance}}{\text{time}}$ • Scalar • m/s +ive	• $\text{Velocity} = \frac{\text{displacement}}{\text{time}}$ • Vector • m/s • +ive -ive 0

Recap 2nd lecture



Lec. 3rd Recap.

Acceleration

⇒ Rate of change of velocity

$$a = \frac{\Delta v}{t} \quad \# \text{ vector} \quad \# \text{ m/s}^2$$

⇒ $a = +ve$, $a = 0$, $a = -ve$

Retardation ♥

Motion

Uniform Motion

$$a = 0$$

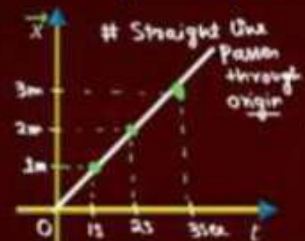
Non-uniform Motion

$$a > 0$$

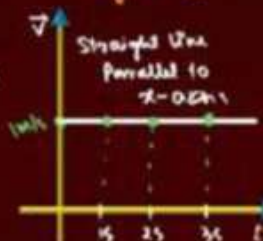
$$a < 0$$

Graphs

① displacement-time graph



② velocity-time

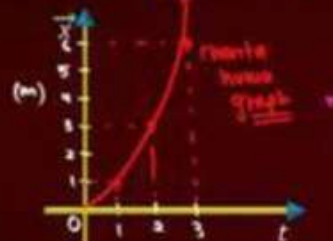


acceleration-time

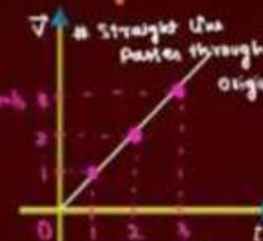


Graphs

① displacement-time graph



② velocity-time

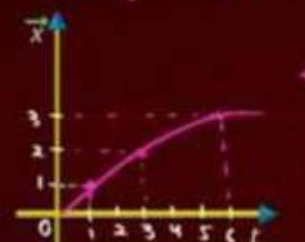


acceleration-time

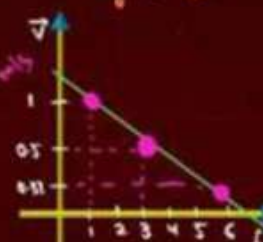


Graphs

① displacement-time graph



② velocity-time



acceleration-time

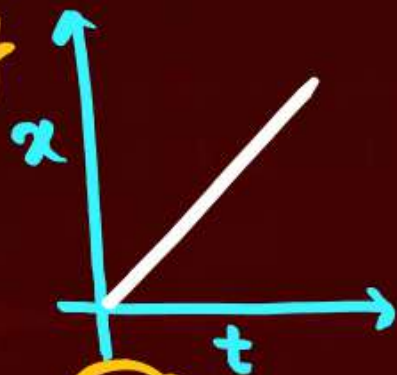


Recap From 4th Lec

→ Board

✓ Slope of $(x)t$ graph gives

Velocity

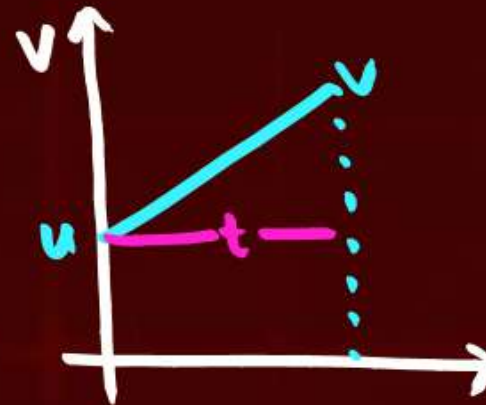


$$v = \frac{dx}{dt}$$

✓ Area of $(v)t$ graph gives
displacement = displacement(x)



Newton equation of motion



$$\textcircled{1} v = u + at$$

$$\textcircled{2} s = ut + \frac{1}{2}at^2$$

$$\textcircled{3} v^2 = u^2 + 2as$$

Acceleration due to gravity (g)

$\textcircled{m}$

↓ g

$$g = -10 \text{ m/s}^2$$

$$g = -9.8 \text{ m/s}^2$$



$$v = u - gt$$

$$s = ut - \frac{1}{2}gt^2$$

$$v^2 = u^2 + 2as$$

Motion in a Straight Line

✓
Frame of reference, Motion in a straight line, Elementary concepts of differentiation and integration for describing motion, uniform and non-uniform motion, and instantaneous velocity, uniformly accelerated motion, velocity - time and position-time graphs. Relations for uniformly accelerated motion (graphical treatment).

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

lecture no. ⑤

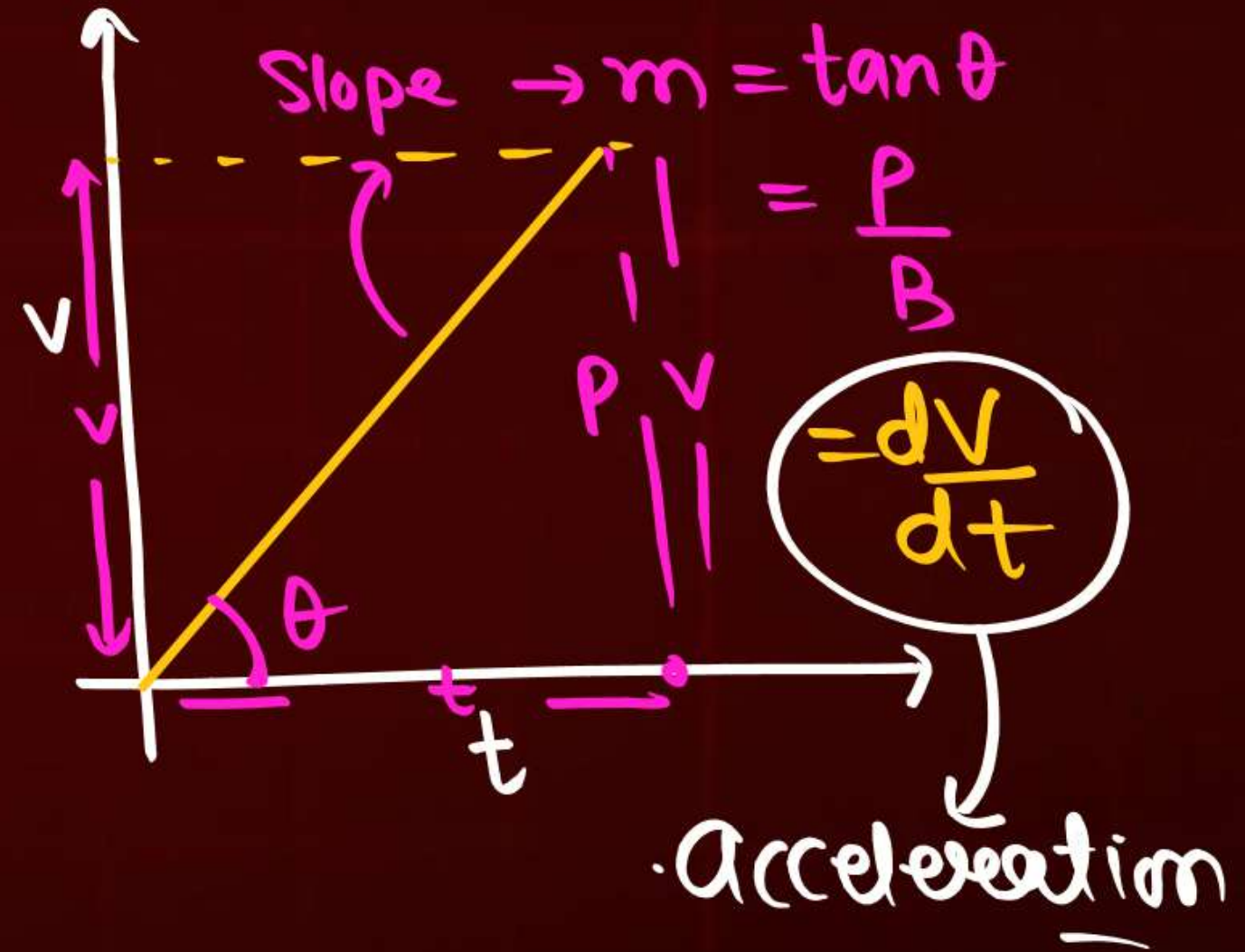
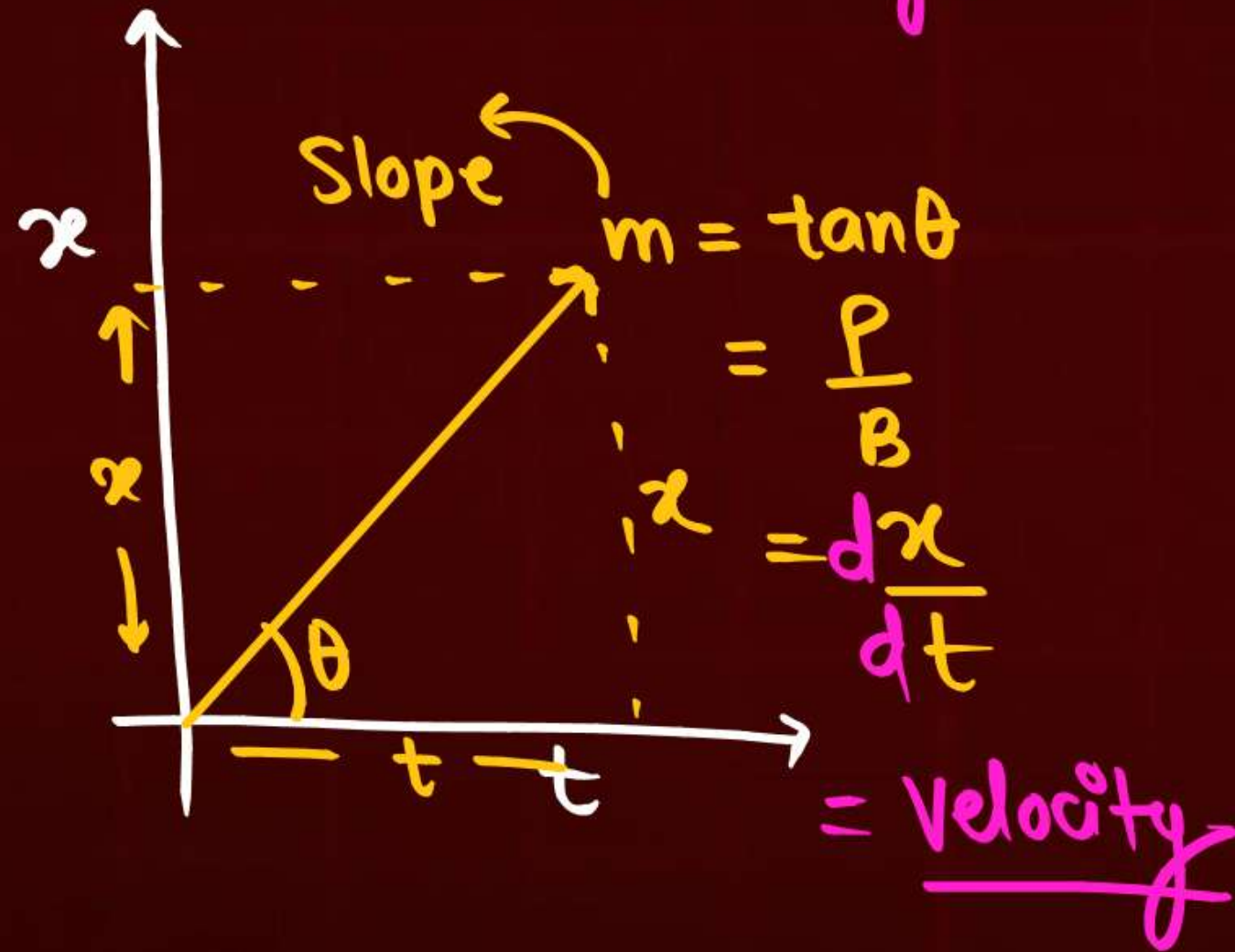
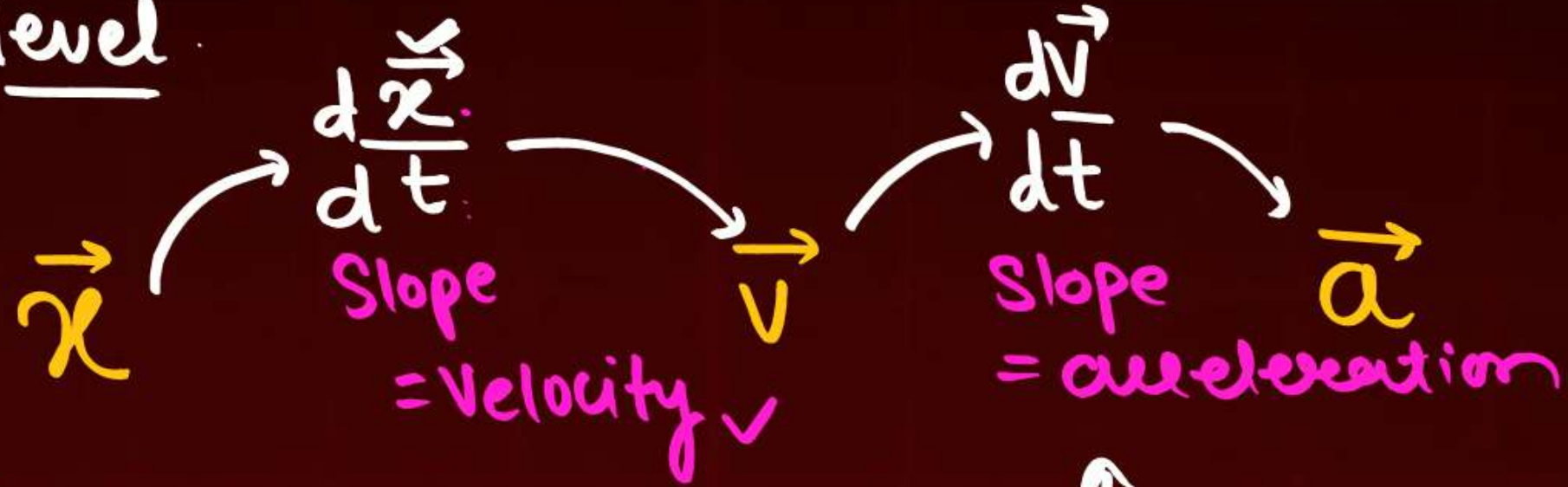


GABBAR

→ Neet ✓
→ Board ✓

NUMERICALS

Neet - level



Q If displacement is the function of t and it is given as $(\underline{x} = 2t^2 + 3t + 2)$ Calculate the $\underline{v}$ at $t = 1\text{sec}$

Sol

$$v = \frac{dx}{dt}$$

$$\frac{dx^n}{dx} = \underline{nx^{n-1}} \quad v = \frac{d(2t^2 + 3t + 2)}{dt}$$

$$v = \frac{d}{dt} 2t^2 + \frac{d}{dt} 3t + \frac{d}{dt} 2$$

$$v = 2 \frac{dt^2}{dt} + 3 \frac{dt}{dt}$$

$$v = 2 \times 2t^{2-1} + 3 \times 1 t^{1-1}$$

$$v = 4t^1 + 3t^0$$

$$v = 4t + 3 \times 1$$

$$v = 4t + 3$$

$$v = 4 \times 1 + 3$$

$$v = 4 + 3 = 7 \text{ m/s.}$$

Q If velocity is the function of (t) and it is given as
 $V = t^2 + t + 5$ Calculate the acc at $t = 2 \text{ sec}$?

Sol

$$a = \frac{dv}{dt}$$

$$a = \frac{d(t^2 + t + 5)}{dt}$$

$$a = \frac{d}{dt} t^2 + \frac{dt^1}{dt} + \frac{d5}{dt} \rightarrow 0$$

$$a = 2t^{2-1} + 1t^{1-1}$$

$$a = 2t^1 + 1 \times t^0$$

$$a = 2t + 1 \times 1$$

$$a = 2t + 1$$

$$a = 2 \times 2 + 1 = \underline{5 \text{ m/s}^2}$$

Neet - level

$$\int a dt = v$$

$$\int v dt = x$$

a

v

x

$$a = \frac{dv}{dt}$$

$$a dt = dv$$

$$\underline{dv} = \underline{a dt}$$

$$\underline{\int dv} = \underline{\int a dt}$$

$$v = \underline{\int a dt}$$

$$v = \frac{dx}{dt}$$

$$v dt = dx$$

$$\int v dt = \int dx$$

$$\underline{\int v dt = x}$$

$$x = \int v dt$$

$x = \text{area under } vt \text{ graph}$

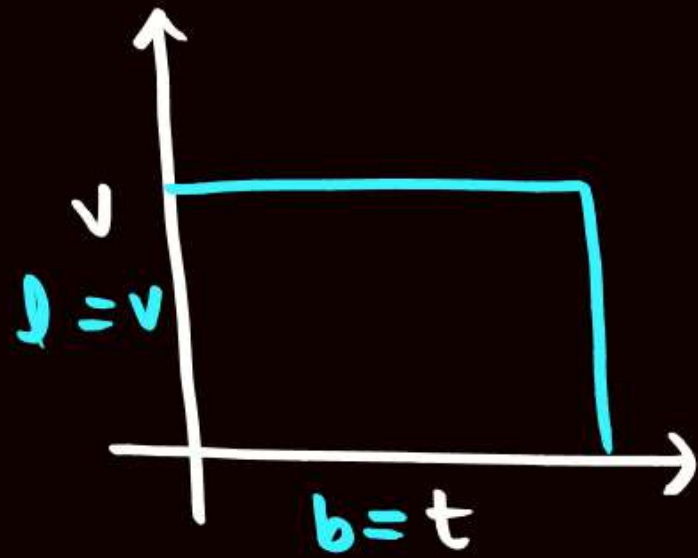
$$v = \int a dt$$

Velocity = area under
at graph

JK BOSE
2024

Neet
2021

⑧

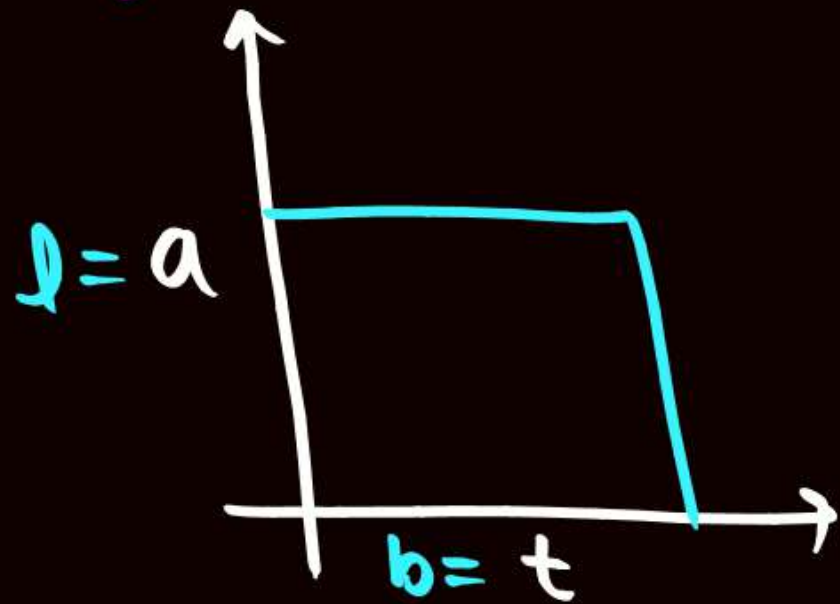


area = $l \times b$
area = $v \times t$
 under $v \times t$
graph area = x ✓

$$v = \frac{x}{t}$$

$$vt = x$$

⑧ J*BoSe
2024



area under at = $l \times b$
graph

area = axt

area = Velocity

$$a = \frac{v}{t}$$

$$axt = v$$

Q. The acc is given as the function of t
Calculate velocity = ? at $t = \underline{2\text{ sec}}$

Sol

$$a = t^2$$

$$v = \int a dt \quad a = t^2$$

$$\int x^n dx$$

$$v = \int t^2 dt$$

$$v = \frac{t^{2+1}}{2+1} = \frac{t^3}{3} = \frac{2^3}{3} = \frac{8}{3} \text{ m/s}$$

JKcet
2012

Q Velocity as the function of time is given by Neet-2017
 $V = 2t$ Calculate the displacement at $t = 2\text{ sec}$?

Sol $x = \int V dt$ $V = 2t$ $\int x^n dx = \frac{x^{n+1}}{n+1}$

$x \rightarrow m$ $x = \int 2t dt$

$v \rightarrow m/s$ $x = 2 \int t dt$

$a \rightarrow m/s^2$ $x = 2 \frac{t^{1+1}}{1+1}$

$x = 2 \frac{t^2}{2}$

$x = 2^2 = \underline{\underline{4m}}$

Question-01

(JK Bosc - 2014)

The slope of velocity-time graph gives:

$$\frac{dv}{dt} = a$$

- A** Velocity
- B** Acceleration ✓
- C** Displacement
- D** Distance

Question-02

If acceleration is zero, motion is:

$$a=0$$

Velocity constt

JKBOSE-2023

- A** Uniform ✓
- B** Circular
- C** Retarded
- D** Non-uniform

Question-03

The SI unit of velocity is:

2019

JKBOSE

A m

B m/s ✓

C m/s²

D km

Question-04

$$x = 2t^2 \quad v = ? \text{ at } t = \underline{2 \text{ s}}$$

If $x = 2t^2$, then velocity at $t = 2\text{s}$ is:

Neet-2021

$$v = \frac{dx}{dt}$$

$$v = \frac{d}{dt} 2t^2$$

$$v = 2 \frac{dt^2}{dt} = 2 \times 2 t^{2-1}$$

$$= 4t = 4 \times 2 = 8 \text{ m/s}$$

A 4 m/s

B 6 m/s

C 8 m/s ✓

D 16 m/s

Question-05JKBOSE - 2017

A body starts from rest. Its initial velocity is:

$$v = 0$$

A 1

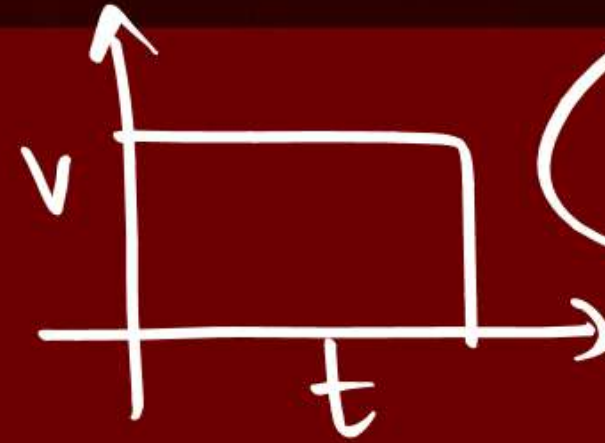
B 0 ✓

C g

D 10

Question-06

JKBOSE
2021



$$\underline{v \times t = x}$$

Area under velocity-time graph gives:

- A** Speed
- B** Acceleration
- C** Displacement ✓
- D** Force

Question-07

2016 JKBOSE

In uniform motion, acceleration is:

$a=0$

- A** Zero ✓
- B** Constant
- C** Increasing
- D** Negative

Question-083mark 2016**Define velocity.**

The velocity is the Rate of change of displacement

$$\vec{V} = \frac{\text{change of displacement}}{\text{time}} = \text{m/s}.$$

Question-09

Define acceleration and write its SI unit. ✓

2019

2mark

Acceleration is rate of change of velocity

$$a = \frac{\Delta v}{t} \Rightarrow \underline{m/s^2}$$

Vector

+ive / -ive / zero

Question-10

Write any two equations of motion.

$$V = u + at$$

$$S = ut + \frac{1}{2}at^2$$

Question-11

What is retardation?

↓ negative acceleration

$$\underline{a = -ve}$$

$$\underline{a = -\text{km/s}^2}$$

Question-12

Derive first equation of motion.

2nd.

3rd

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from graphical method

24

23

22

21

20

19

18

Question-14

A body starts from rest and accelerates at 3 m/s^2 for 4 s. Find final velocity.

Sol

$$u = 0$$
$$a = 3 \text{ m/s}^2$$
$$t = 4 \text{ sec}$$
$$v = ?$$

$$\textcircled{1} v = u + at \checkmark$$
$$v = 0 + 3 \times 4$$
$$v = \underline{12 \text{ m/s}}$$

$$\textcircled{3} v^2 = u^2 + 2as \checkmark$$

JKcet
2013

Question-15

JKBOSE-2019



A car moving with 8 m/s accelerates at 2 m/s² for 5 s. Find displacement.

$$S = ?$$

$$u = 8 \checkmark$$

$$a = 2 \checkmark$$

$$t = 5 \text{ sec} \checkmark$$

$$\textcircled{1} v = u + at \quad X$$

$$\textcircled{2} S = ut + \frac{1}{2} at^2$$

$$S = 8 \times 5 + \frac{1}{2} \cancel{2} \times (5)^2$$

$$S = 40 + 25 = \underline{65 \text{ m}} \checkmark$$

Question-16

→ JKBOSE 2018



A body moving with 5 m/s accelerates at 2 m/s² for 3 s. Find final velocity and displacement.

$$v = ?$$

$$s = ?$$

$$u = 5 \text{ m/s}$$

$$a = 2 \text{ m/s}^2$$

$$t = \underline{3 \text{ s}}$$

$$\textcircled{1} v = u + at$$

$$v = 5 + 2 \times 3$$

$$v = 5 + 6$$

$$v = 11 \text{ m/s}$$

$$\begin{aligned} S &= ut + \frac{1}{2} at^2 \\ &= 5 \times 3 + \frac{1}{2} \times 2 \times 3^2 \\ &= 15 + 9 \\ &= \underline{24 \text{ m}} \end{aligned}$$

Question-17

$$a = \frac{v - u}{t}$$

A car increases its velocity from 10 m/s to 30 m/s in 5 s.

Find:

- (i) Acceleration ✓
- (ii) Distance travelled ✓

$$a = 4 \text{ m/s}^2$$

(i) $u = 10$

$v = 30$

$t = 5 \text{ sec}$

$a = ?$

$$v = u + at$$

$$30 = 10 + a \times 5$$

$$30 - 10 = a \times 5$$

$$20 = a \times 5$$

$$4 \text{ m/s}^2 = \frac{20}{5} = a$$

$$S = ut + \frac{1}{2} at^2$$

$$S = 10 \times 5 + \frac{1}{2} \times 4 \times 5^2$$

$$S = 50 + 2 \times 25$$

$$S = 50 + 50 = 100 \text{ m} \checkmark$$

Question-18

Rs-100

A body moves with an initial velocity of 12 m/s and accelerates uniformly at 3 m/s^2 for 6 s .

Find:

- (i) Final velocity
- (ii) Displacement covered

HOMework

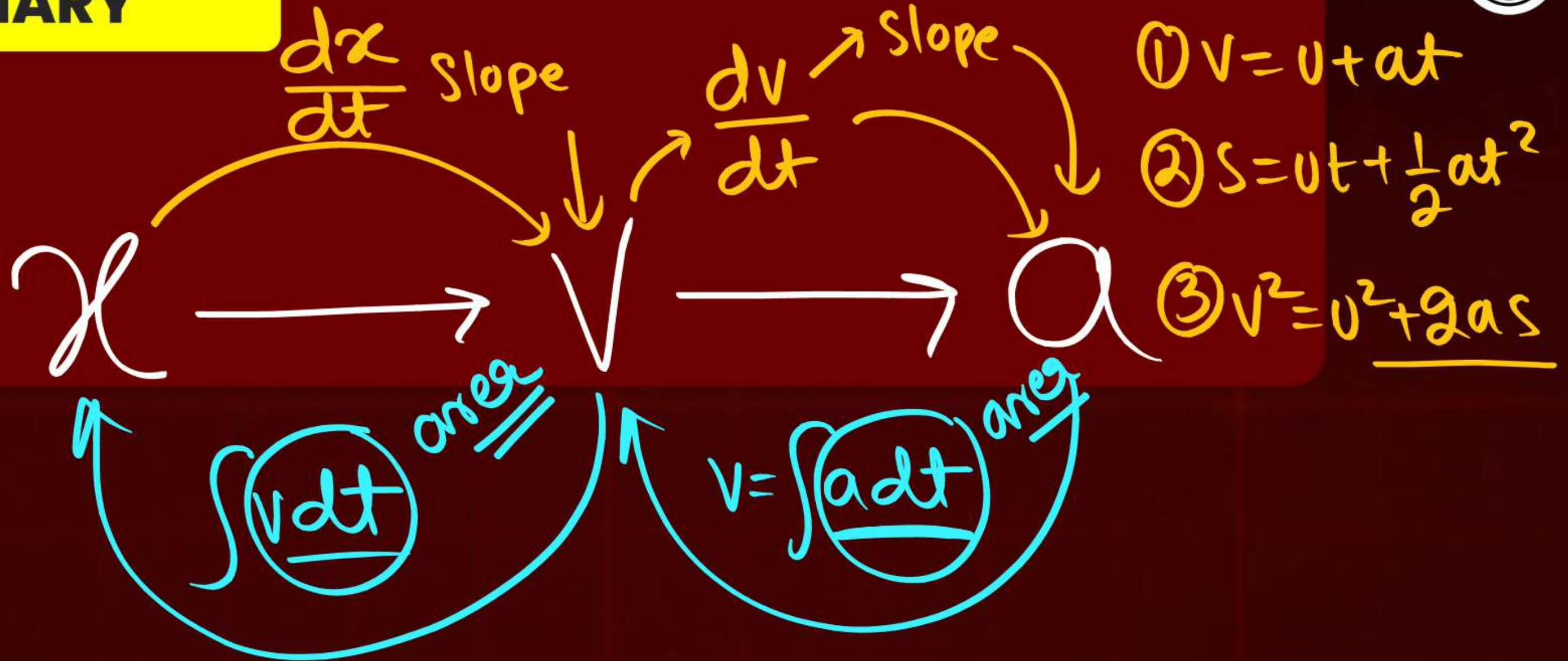
Q $V = 2t^2 + t + 3$
at $t = 1 \text{ sec}$.
Calculate $x = ?$

$R_s = 50$

Q $a = 3t^2 + t + 5$
at $t = 2 \text{ sec}$

Calculate $v = ?$

SUMMARY



Thank
You